

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $10y^2 + 12y =$

(2) Factor: $y^2 + 9y + 18 =$

(3) Factor: $16y^2 + 8y =$

(4) Factor: $6x^2 + 15x =$

(5) Factor: $x^2 + 2x + 1 =$

(6) Factor: $x^2 - 36 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $y^2 - 64 =$

(2) Factor: $10x^2 + 8x =$

(3) Factor: $x^2 - 9 =$

(4) Factor: $12y^2 + 4y =$

(5) Factor: $4x^2 + 12x =$

(6) Factor: $x^2 - 25 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $x^2 + 8x + 16 =$

(2) Factor: $16x^2 + 4x =$

(3) Factor: $y^2 + 8y + 12 =$

(4) Factor: $y^2 + 5y + 6 =$

(5) Factor: $x^2 - 25 =$

(6) Factor: $y^2 - 36 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $x^2 - 16 =$

(2) Factor: $y^2 + 7y + 12 =$

(3) Factor: $y^2 - 64 =$

(4) Factor: $14x^2 + 8x =$

(5) Factor: $24y^2 + 12y =$

(6) Factor: $24x^2 + 6x =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $28x^2 + 20x =$

(2) Factor: $x^2 - 49 =$

(3) Factor: $y^2 + 9y + 18 =$

(4) Factor: $y^2 - 16 =$

(5) Factor: $y^2 - 16 =$

(6) Factor: $y^2 + 4y + 4 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $12x^2 + 24x =$

(2) Factor: $x^2 + 6x + 8 =$

(3) Factor: $y^2 - 4 =$

(4) Factor: $28x^2 + 8x =$

(5) Factor: $x^2 - 25 =$

(6) Factor: $x^2 - 49 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $y^2 - 81 =$

(2) Factor: $y^2 + 6y + 9 =$

(3) Factor: $y^2 - 81 =$

(4) Factor: $16y^2 + 8y =$

(5) Factor: $y^2 - 64 =$

(6) Factor: $14x^2 + 6x =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $x^2 - 81 =$

(2) Factor: $20y^2 + 24y =$

(3) Factor: $6y^2 + 8y =$

(4) Factor: $x^2 - 81 =$

(5) Factor: $x^2 - 25 =$

(6) Factor: $20y^2 + 24y =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $y^2 - 36 =$

(2) Factor: $12x^2 + 9x =$

(3) Factor: $x^2 + 9x + 20 =$

(4) Factor: $y^2 + 7y + 12 =$

(5) Factor: $y^2 + 6y + 8 =$

(6) Factor: $y^2 + 4y + 4 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $8y^2 + 8y =$

(2) Factor: $y^2 + 4y + 3 =$

(3) Factor: $y^2 + 3y + 2 =$

(4) Factor: $12x^2 + 12x =$

(5) Factor: $y^2 + 6y + 9 =$

(6) Factor: $x^2 - 49 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $y^2 - 25 =$

(2) Factor: $x^2 + 9x + 18 =$

(3) Factor: $6x^2 + 15x =$

(4) Factor: $y^2 - 4 =$

(5) Factor: $x^2 + 3x + 2 =$

(6) Factor: $y^2 - 49 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $y^2 + 7y + 10 =$

(2) Factor: $x^2 + 2x + 1 =$

(3) Factor: $x^2 + 5x + 4 =$

(4) Factor: $4x^2 + 6x =$

(5) Factor: $16y^2 + 8y =$

(6) Factor: $y^2 + 5y + 4 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $x^2 + 3x + 2 =$

(2) Factor: $8y^2 + 4y =$

(3) Factor: $y^2 + 3y + 2 =$

(4) Factor: $x^2 - 25 =$

(5) Factor: $x^2 + 5x + 4 =$

(6) Factor: $x^2 - 25 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $18y^2 + 15y =$

(2) Factor: $6y^2 + 18y =$

(3) Factor: $4x^2 + 8x =$

(4) Factor: $x^2 - 64 =$

(5) Factor: $y^2 - 9 =$

(6) Factor: $12x^2 + 15x =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $x^2 - 16 =$

(2) Factor: $20x^2 + 20x =$

(3) Factor: $y^2 + 2y + 1 =$

(4) Factor: $24y^2 + 8y =$

(5) Factor: $y^2 + 5y + 4 =$

(6) Factor: $y^2 - 64 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $y^2 + 7y + 12 =$

(2) Factor: $x^2 + 6x + 5 =$

(3) Factor: $y^2 + 2y + 1 =$

(4) Factor: $x^2 - 9 =$

(5) Factor: $y^2 - 49 =$

(6) Factor: $24y^2 + 15y =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $12x^2 + 12x =$

(2) Factor: $y^2 + 4y + 4 =$

(3) Factor: $y^2 - 36 =$

(4) Factor: $x^2 - 9 =$

(5) Factor: $12y^2 + 15y =$

(6) Factor: $x^2 - 49 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $y^2 - 16 =$

(2) Factor: $y^2 - 36 =$

(3) Factor: $y^2 + 5y + 6 =$

(4) Factor: $y^2 - 64 =$

(5) Factor: $x^2 - 25 =$

(6) Factor: $x^2 + 7x + 6 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $x^2 - 81 =$

(2) Factor: $x^2 - 64 =$

(3) Factor: $y^2 + 8y + 12 =$

(4) Factor: $x^2 - 36 =$

(5) Factor: $x^2 + 8x + 16 =$

(6) Factor: $24y^2 + 20y =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $x^2 - 81 =$

(2) Factor: $x^2 + 5x + 4 =$

(3) Factor: $6y^2 + 15y =$

(4) Factor: $y^2 + 7y + 12 =$

(5) Factor: $y^2 - 16 =$

(6) Factor: $y^2 - 25 =$

Answer Key

JPMethod Polynomials

F41a

- (1) $2y(5y + 6)$
- (2) $(y + 3)(y + 6)$
- (3) $4y(4y + 2)$
- (4) $3x(2x + 5)$
- (5) $(x + 1)(x + 1)$
- (6) $(x + 6)(x - 6)$

F42a

- (1) $(y + 8)(y - 8)$
- (2) $2x(5x + 4)$
- (3) $(x + 3)(x - 3)$
- (4) $2y(6y + 2)$
- (5) $2x(2x + 6)$
- (6) $(x + 5)(x - 5)$

F41b

- (1) $(x + 4)(x - 4)$
- (2) $(y + 3)(y + 4)$
- (3) $(y + 8)(y - 8)$
- (4) $2x(7x + 4)$
- (5) $4y(6y + 3)$
- (6) $3x(8x + 2)$

F42b

- (1) $(x + 4)(x + 4)$
- (2) $2x(8x + 2)$
- (3) $(y + 2)(y + 6)$
- (4) $(y + 3)(y + 2)$
- (5) $(x + 5)(x - 5)$
- (6) $(y + 6)(y - 6)$

Answer Key

JPMethod Polynomials

F45a

- (1) $(y + 6)(y - 6)$
- (2) $3x(4x + 3)$
- (3) $(x + 4)(x + 5)$
- (4) $(y + 3)(y + 4)$
- (5) $(y + 4)(y + 2)$
- (6) $(y + 2)(y + 2)$

F46a

- (1) $4y(2y + 2)$
- (2) $(y + 3)(y + 1)$
- (3) $(y + 2)(y + 1)$
- (4) $4x(3x + 3)$
- (5) $(y + 3)(y + 3)$
- (6) $(x + 7)(x - 7)$

F45b

- (1) $(y + 2)(y + 5)$
- (2) $(x + 1)(x + 1)$
- (3) $(x + 1)(x + 4)$
- (4) $2x(2x + 3)$
- (5) $4y(4y + 2)$
- (6) $(y + 4)(y + 1)$

F46b

- (1) $(y + 5)(y - 5)$
- (2) $(x + 3)(x + 6)$
- (3) $3x(2x + 5)$
- (4) $(y + 2)(y - 2)$
- (5) $(x + 2)(x + 1)$
- (6) $(y + 7)(y - 7)$

Answer Key

JPMethod Polynomials

F47a

- (1) $(x + 1)(x + 2)$
- (2) $2y(4y + 2)$
- (3) $(y + 1)(y + 2)$
- (4) $(x + 5)(x - 5)$
- (5) $(x + 4)(x + 1)$
- (6) $(x + 5)(x - 5)$

F48a

- (1) $3y(6y + 5)$
- (2) $3y(2y + 6)$
- (3) $2x(2x + 4)$
- (4) $(x + 8)(x - 8)$
- (5) $(y + 3)(y - 3)$
- (6) $3x(4x + 5)$

F47b

- (1) $(y + 3)(y + 4)$
- (2) $(x + 1)(x + 5)$
- (3) $(y + 1)(y + 1)$
- (4) $(x + 3)(x - 3)$
- (5) $(y + 7)(y - 7)$
- (6) $3y(8y + 5)$

F48b

- (1) $(x + 4)(x - 4)$
- (2) $4x(5x + 5)$
- (3) $(y + 1)(y + 1)$
- (4) $4y(6y + 2)$
- (5) $(y + 4)(y + 1)$
- (6) $(y + 8)(y - 8)$

Answer Key

JPMethod Polynomials

F43a

- (1) $4x(7x + 5)$
- (2) $(x + 7)(x - 7)$
- (3) $(y + 3)(y + 6)$
- (4) $(y + 4)(y - 4)$
- (5) $(y + 4)(y - 4)$
- (6) $(y + 2)(y + 2)$

F44a

- (1) $4x(3x + 6)$
- (2) $(x + 4)(x + 2)$
- (3) $(y + 2)(y - 2)$
- (4) $4x(7x + 2)$
- (5) $(x + 5)(x - 5)$
- (6) $(x + 7)(x - 7)$

F43b

- (1) $(x + 9)(x - 9)$
- (2) $4y(5y + 6)$
- (3) $2y(3y + 4)$
- (4) $(x + 9)(x - 9)$
- (5) $(x + 5)(x - 5)$
- (6) $4y(5y + 6)$

F44b

- (1) $(y + 9)(y - 9)$
- (2) $(y + 3)(y + 3)$
- (3) $(y + 9)(y - 9)$
- (4) $2y(8y + 4)$
- (5) $(y + 8)(y - 8)$
- (6) $2x(7x + 3)$

F49a

- (1) $2x(6x + 6)$
- (2) $(y + 2)(y + 2)$
- (3) $(y + 6)(y - 6)$
- (4) $(x + 3)(x - 3)$
- (5) $3y(4y + 5)$
- (6) $(x + 7)(x - 7)$

F50a

- (1) $(y + 4)(y - 4)$
- (2) $(y + 6)(y - 6)$
- (3) $(y + 2)(y + 3)$
- (4) $(y + 8)(y - 8)$
- (5) $(x + 5)(x - 5)$
- (6) $(x + 1)(x + 6)$

F49b

- (1) $(x + 9)(x - 9)$
- (2) $(x + 4)(x + 1)$
- (3) $3y(2y + 5)$
- (4) $(y + 3)(y + 4)$
- (5) $(y + 4)(y - 4)$
- (6) $(y + 5)(y - 5)$

F50b

- (1) $(x + 9)(x - 9)$
- (2) $(x + 8)(x - 8)$
- (3) $(y + 2)(y + 6)$
- (4) $(x + 6)(x - 6)$
- (5) $(x + 4)(x + 4)$
- (6) $4y(6y + 5)$

